EXERCISES

(1) What does each of the following commands do? Print the output and attach it as part of your Lab3 document

(a) cd / to root directory
(b) cd ~ to home directory
(c) cd to home directory
(d) cd .. to parent directory
(e) cd . To current directory
(f) cd – to previous directory

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd /
ibab@IBAB-MSC-BDB-comp16-ibab:/$ cd ~
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd ..
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ cd .
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ cd -
/home
ibab@IBAB-MSC-BDB-comp16-ibab:/home$
```

(2) When using the ls command, what is the difference between -R option and -r option? Demonstrate the difference by typing the appropriate commands and printing the output to your Lab3 document.

ls -R : Recursively subdirectories listing
ls -r : reverse order while sorting

```
ibab@IBAB-MSC-BDB-comp16-ibab: /home
/home
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -r
workshop  newuser  lost+found  khan  ibab
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -R
.:
ibab  khan  lost+found  newuser  workshop

./ibab:
 22         23sym       Downloads   lab3      Music      snap        Videos
 23         Desktop     ibab        ls        Pictures   Templates
 23_slink   Documents   igv        'MEGA X'   Public     testfile

./ibab/Desktop:
025

./ibab/Desktop/025:
BDBP106_Lab2.odt  BDBP106_Lab2.pdf  Lab1LPPBDBP106.odt  Lab1LPPBDBP106.pdf

./ibab/Documents:

./ibab/Downloads:
22  lab2.pdf

./ibab/Downloads/22:

./ibab/ibab:

./ibab/igv:
genomes  igv0.log  prefs.properties

./ibab/igv/genomes:

./ibab/lab3:

'./ibab/MEGA X':
Examples

'./ibab/MEGA X/Examples':
ABI01.abi                    mtCDNACalibrationDensities.txt
Chloroplast_Martin.meg       mtCDNACalibration.txt
command_statements_dists.meg mtCDNA.meg
command_statements.meg       mtCDNA.nwk
Contigs.meg                  mtCDNAOutgroup.txt
Crab_rRNA.meg                NeiKumar2000
Distance_Data.meg            NewickTree.nwk
D-loop_Vigilant.meg          reltime_with_dated_tips_RTDT_alignment.meg
Drosophila_Adh.meg           reltime_with_dated_tips_RTDT_outgroup.txt
gene_tree.nwk                reltime_with_dated_tips_RTDT_sample_times.txt
HLA-3Seq.meg                 reltime_with_dated_tips_RTDT_tree.nwk
hsp20.fas                    species_tree.nwk
hsp20.meg                    taxa_to_species_map.txt
Hum_Dist.meg

'./ibab/MEGA X/Examples/NeiKumar2000':
Pg024_Chap_02_Exp_01_Data.meg  Pg174_Chap_09_Exp_01_Data.meg
Pg027_Chap_02_Exp_02_Data.meg  Pg183_Chap_09_Fig_04_Data.meg
```

(3) How do I find out what directory I am currently in?
pwd-present working directory

```
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ pwd
/home
ibab@IBAB-MSC-BDB-comp16-ibab:/home$
```

(4) How do I list files sorted by modification time, but with oldest first? Find the correct combination of options to do this task. Print the output of your command with appropriate screenshots to demonstrate this in your Lab3 document.

```
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -ltr
total 32
drwx------  2 root     root     16384 Jul 24  2023 lost+found
drwxr-x--- 16 khan     khan      4096 Aug  3  2023 khan
drwxr-x--- 20 workshop workshop  4096 Aug  4  2023 workshop
drwxr-x---  2     1003     1003  4096 Aug 14  2023 newuser
drwxr-x--- 26 ibab     ibab      4096 Jul 24 14:14 ibab
ibab@IBAB-MSC-BDB-comp16-ibab:/home$
```

(5) What is the difference between (explain in your own words) ls -ltr and ls -l? Demonstrate the difference with appropriate screenshot and explanations.
ls-l long listing format
ls -ltr long listing time format in reverse

```
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -ltr
total 32
drwx------  2 root     root     16384 Jul 24  2023 lost+found
drwxr-x--- 16 khan     khan      4096 Aug  3  2023 khan
drwxr-x--- 20 workshop workshop  4096 Aug  4  2023 workshop
drwxr-x---  2     1003     1003  4096 Aug 14  2023 newuser
drwxr-x--- 26 ibab     ibab      4096 Jul 24 14:14 ibab
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -l
total 32
drwxr-x--- 26 ibab     ibab      4096 Jul 24 14:14 ibab
drwxr-x--- 16 khan     khan      4096 Aug  3  2023 khan
drwx------  2 root     root     16384 Jul 24  2023 lost+found
drwxr-x---  2     1003     1003  4096 Aug 14  2023 newuser
drwxr-x--- 20 workshop workshop  4096 Aug  4  2023 workshop
ibab@IBAB-MSC-BDB-comp16-ibab:/home$
```

(6) What is the difference between ls -alF, ls -A and ls -CF. Demonstrate the differences with appropriate examples as screenshot and explanations.
ls-alF : all list file format
ls -A :Author
ls – CF : Color and file type

```
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -alF
total 40
drwxr-xr-x  7 root     root      4096 Jul 15 15:01 ./
drwxr-xr-x 20 root     root      4096 Jul 24  2023 ../
drwxr-x--- 26 ibab     ibab      4096 Jul 24 14:14 ibab/
drwxr-x--- 16 khan     khan      4096 Aug  3  2023 khan/
drwx------  2 root     root     16384 Jul 24  2023 lost+found/
drwxr-x---  2     1003     1003  4096 Aug 14  2023 newuser/
drwxr-x--- 20 workshop workshop  4096 Aug  4  2023 workshop/
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -A
ibab  khan  lost+found  newuser  workshop
ibab@IBAB-MSC-BDB-comp16-ibab:/home$ ls -CF
ibab/  khan/  lost+found/  newuser/  workshop/
ibab@IBAB-MSC-BDB-comp16-ibab:/home$
```

(7) Describe how you would use a combination of color codes in the command prompt appearance. Demonstrate this using an example where you managed to combine the colors blue and green?

```
ibab@IBAB-MSC-BDB-comp16-ibab:/$ export PS1="\e[0;34;32m[u@\h\W]\$\e[0m"
[u@IBAB-MSC-BDB-comp16-ibab/]$
```

(8) Find the option of ls that lists the contents of a directory in the long format, but without the primary group information. (Note: ls -l lists files in the long format with the primary group information)

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ ls -lG
total 60
-rw-rw-r-- 1 ibab    0 Jul 23 09:08  22
-rw-rw-r-- 1 ibab    0 Jul 23 09:14  23
lrwxrwxrwx 1 ibab    2 Jul 23 09:29  23_slink -> 23
lrwxrwxrwx 1 ibab    2 Jul 23 09:28  23sym -> 23
drwxr-xr-x 3 ibab 4096 Jul 17 16:29  Desktop
drwxr-xr-x 2 ibab 4096 Jul 16 16:21  Documents
drwxr-xr-x 4 ibab 4096 Jul 24 14:35  Downloads
drwxrwxr-x 2 ibab 4096 Jul 17 11:16  ibab
drwxrwxr-x 3 ibab 4096 Jul 17 14:59  igv
drwxrwxr-x 2 ibab 4096 Jul 24 14:14  lab3
-rw-r--r-- 1 ibab 2275 Jul 21 11:10  ls
drwxrwxr-x 3 ibab 4096 Jul 17 15:06 'MEGA X'
drwxr-xr-x 2 ibab 4096 Jul 16 16:21  Music
drwxrwxr-x 3 ibab 4096 Jul 23 09:20  Pictures
drwxr-xr-x 2 ibab 4096 Jul 16 16:21  Public
drwx------ 4 ibab 4096 Jul 17 11:12  snap
drwxr-xr-x 2 ibab 4096 Jul 16 16:21  Templates
drwxrwxr-x 2 ibab 4096 Jul 21 11:43  testfile
drwxr-xr-x 2 ibab 4096 Jul 16 16:21  Videos
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(9) Download the etc.zip folder sent to you by email, unzip it and rename it as etc_learn. Change your directory to etc_learn. In this directory do the following using the necessary wildcards:

(a) list the files or directories starting with b

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd Downloads/
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cd etc_learn/
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ ls b*
bash.bashrc  bash_completion  bindresvport.blacklist  brlapi.key  brltty.conf

bash_completion.d:

binfmt.d:

bluetooth:

brltty:

byobu:
```

(b) list the files or directories starting with any of the letters from a-k

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ ls [a-k]*
adduser.conf            crypttab        fuse.conf     hosts
anacrontab              debconf.conf    gai.conf      hosts.allow
apg.conf                debian_version  gprofng.rc    hosts.deny
bash.bashrc             deluser.conf    group         inputrc
bash_completion         dhcpcd.conf     group-        issue
bindresvport.blacklist  e2scrub.conf    gshadow       issue.net
brlapi.key              environment     gshadow-      kerneloops.conf
brltty.conf             ethertypes      hdparm.conf
ca-certificates.conf    fprintd.conf    host.conf
crontab                 fstab           hostname

alsa:

alternatives:

apm:

apparmor:

apparmor.d:

apport:

apt:

avahi:

bash_completion.d:

binfmt.d:

bluetooth:

brltty:

byobu:

ca-certificates:

chatscripts:

cloud:

colord:

console-setup:

cracklib:

credstore:
```

(c) list files or directories ending with d

```
dhcp:
dictionaries-common:
dpkg:
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ ls *d
machine-id
apparmor.d:
bash_completion.d:
binfmt.d:
cloud:
colord:
credstore.encrypted:
cron.d:
debuginfod:
depmod.d:
environment.d:
fwupd:
glvnd:
grub.d:
ifplugd:
init.d:
insserv.conf.d:
krb5.conf.d:
ld.so.conf.d:
libibverbs.d:
libpaper.d:
logrotate.d:
modprobe.d:
modules-load.d:
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$
```

(d) list the files or directories containing the pattern with conf. How many are there?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ ls *conf.*
debconf.conf

insserv.conf.d:

krb5.conf.d:

ld.so.conf.d:
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$
```

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(e) Output the last three lines of the file /etc/resolve.conf

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ tail -n 3 /etc/resolv.conf
nameserver 127.0.0.53
options edns0 trust-ad
search ibab.ac.in
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$
```

(f) list files that begin with g or l

```
gtk-3.0:
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$ ls [g,l]*
gai.conf  gprofng.rc  group  group-  gshadow  gshadow-  ld.so.cache  ld.so.conf  legal  libao.conf  libaudit.conf  locale.alias  locale.conf  locale.gen  login.defs  logrotate.conf  lsb-release

gdb:

gdm3:

geoclue:

ghostscript:

glvnd:

gnome:

gnome-remote-desktop:

gnutls:

groff:

grub.d:

gss:

gtk-2.0:

gtk-3.0:

landscape:

ldap:

ld.so.conf.d:

libblockdev:

libibverbs.d:

libnl-3:

libpaper.d:

libreoffice:

logcheck:

logrotate.d:

lvm:
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/etc_learn$
```

(10) Create the following directories: /home/ibab/lab3/Q10/q1 and /home/ibab/lab3/Q10/q2 using one command. Attach a screenshot of the commands and the outputs of this and below exercises in your document.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ mkdir --p home/ibab/lab3/Q10/q1 home/ibab/lab3/Q10/q2
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$
```

(i) Enter the q1 directory using one command

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ cd q1
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10/q1$
```

(ii) Change directories from q1 to q2 using one command

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ cd /home/ibab/ibab/lab3/Q10/q2
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10/q2$
```

(iii) copy the q1 directory to q1_copy

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ cp -r q1 q1_copy
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ pwd
/home/ibab/ibab/lab3/Q10
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ ls -l
total 12
drwxrwxr-x 2 ibab ibab 4096 Jul 24 16:35 q1
drwxrwxr-x 2 ibab ibab 4096 Jul 24 16:50 q1_copy
drwxrwxr-x 2 ibab ibab 4096 Jul 24 16:37 q2
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$
```

(iv) delete the q1_copy interactively

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ rm -ri q1_copy
rm: remove directory 'q1_copy'? y
ibab@IBAB-MSC-BDB-comp16-ibab:~/ibab/lab3/Q10$ ls
q1  q2
```